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Emotion recognition in Hindi text using multilingual BERT transformer

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Abstract

Emotions are a vital and fundamental part of our existence. Whatever we do, say, or do not say somehow reflects our feelings, however not immediately. To comprehend human's most fundamental behaviour, we must examine these feelings using emotional data. According to the extensive literature review, categorising speech text into multiple classes is now undergoing extensive investigation. The application of this research is very limited in local and regional languages such as Hindi. This study focuses on text emotion analysis, specifically for the Hindi language. In our study, BHAAV Dataset is used, which consists of 20,304 sentences, where every other sentence has been manually annotated into one of the five emotion categories (Anger, Suspense, Joy, Sad, Neutral). Comparison of multiple machine learning and deep learning techniques with word embedding is used to demonstrate accuracy. And then, the trained model is used to predict the emotions of Hindi text. The best performance were observed in case of mBERT model with loss- 0.1689 ,balanced_accuracy- 93.88%, recall- 93.44%, auc- 99.55% and precision- 94.39 % on training data, while loss- 0.3073, balanced_accuracy- 91.84%, recall- 91.74%, auc- 98.46% and precision- 92.01% on testing data.

Keywords BERT transformer · Deep learning · Emotion recognition · Hindi text · Machine learning · Sentiment analysis · Text analysis

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1 Introduction

Language is a very effective tool for communicating and conveying information. It serves as a technique for expressing one's feelings. Since its outset in 1950 and rebirth in the twentieth century, Artificial intelligence has made essential contributions in various sectors, including natural language processing (NLP), by providing acceptable answers for real-world personal and societal challenges. The NLP uses computational and linguistic techniques to help machines understand and occasionally produce linguistic information in the form of audio and text [42]. Sentiment analysis (SA), translation systems, questions and answers systems, information retrieval (IR), text generation system, and others are notable accomplishments in the field of NLP under active investigation [9]. Emotion detection (ED) is an area of sentiment analysis that aims to extract finer-grained emotions such as joyful, sad, furious etc., from human languages rather than coarse-grained and generic polarity assignments. Emotion Detection is the process of determining the emotion behind any text or voice and is a more advanced form of sentiment analysis (SA). As a result, ED is a synergistic combination of emotions. Emotions can be discrete or multidimensional. ED is also known as effective computing [10], and it derives its core from the use of emotion-based technologies in a variety of industries in order to facilitate smooth decision making. Table 1 differentiates the concept of Emotion detection with sentiment analysis.

The inspiration for this study stems from the current surge of interest in emotion analysis. The use of technology to help individuals identify emotions is a relatively recent research subject. In general, the technology works best when several modalities are combined. The majority of research has been on automating the recognition of facial expressions in video and voice. More of the current research in this domain has been on recognising the polarity of emotion (positive/negative). Identifying varieties of emotions and their intensity — the subject of this work – is one of the least investigated sentiment domains. Another issue with Emotion Recognition Models is that most of them are only available in English. As a result, emotion identification in languages other than English, particularly regional and local languages, presents a significant endeavour and potential for academics.

This study aims to employ Natural Language Processing techniques to recognise feelings and emotions in Hindi literature.The novelty of this approach lies in the efficiency and accuracy of emotion detection for the Hindi language.

Table 1 Sentiment analysis vs emotion detection

S. No.	Sentiment analysis	Emotion detection
1	Sentiment refers to assessments, which are often positive, negative, or neutral	Emotion recognition would try to figure out how furious, pleased, or sad the user is
2	The most typical text categorisation method examines an incoming message and informs the user's intent	The cognitive and behavioural tactics that people employ to affect their own emotional experience are called emotion recognition
3	Textual data is the most commonly used source for sentiment analysis	Emotion identification extends beyond text, for example, to facial expressions
4	It may be used in product reviews to determine whether or not the reviewer enjoyed the product	Typically is used for conversational data, such as discussions with chatbots

Another distinctive characteristic of this research is that we are able to solve the problem of Word Sense Disambiguation and incorporate contextual meaning for better emotion prediction. Through this research, we also concluded that word embedding played a significant role in increasing the accuracy of emotion recognition.

2 Related work

Manshu et al. in a paper [28], suggest a prior-knowledge information hierarchical attention network (HANP) for the cross-domain sentiment classification project. The studies are carried out using Amazon reviews datasets [6], which have been widely utilised for Cross-Domain categorisation tasks. The emotion dictionary match layer may additionally capture crucial pivots, non-pivots, and dis-pivots. The HANP can determine the meaning of a non-pivot or dis-pivot by searching for synonyms. Experiments on the Amazon review dataset demonstrate the efficacy of HANP.

Huang et al. [18] investigated the performance of BERT. They altered BERT and produced two models, namely FriendsBERT and ChatBERT, using distinct pre-training tasks before the formal training procedure for the Friends and EmotionPush datasets. EmotionLines [13] is a conversation dataset divided into two subgroups based on the source of the dialogues: Friends and EmotionPush. The former is based on the scripts of the Friends television series. The other consists of Facebook Messenger talks. Each subset contains 1,000 English conversations, which may be further subdivided into a few successive utterances. There were 240 conversations in the testing dataset, with 3296 and 3536 utterances in Friends and EmotionPush. On the Friends and EmotionPush testing datasets, the provided technique gets 81.5% and 88.5% micro F1-scores, respectively.

Huang et al. [19] presented an HRLCE and BERT model for text-based emotion detection on Twitter' Tweets. In more technical terms, the challenge is to identify the emotion of a user's speech as Happy, Sad, Angry, or Others given a textual user utterance and two turns of context in a discussion. They suggested an ensemble technique comprised of two deep learning models, the Hierarchical LSTMs for Contextual Emotion Detection (HRLCE) and the BERT. The tweets are collected from Twitter and annotated in four categories: happy, sad, angry, and others. The dataset contains 30160 sentences in train data and 5509 in test data. The model evaluation on the test dataset yielded a 0.7709 harmonic mean of the macro-F1 scores across the Happy, Angry, and Sad categories.

Ragheb et al. [36] presented an attention-based model for categorizing emotions. The suggested model employs transfer learning via universal language modelling, consisting of successive layers of Bi-directional Long Term Short Term Memory (Bi-LSTM) units. The model is made up of two major steps: encoder and classifier. Each discussion is a three-person chat. The labels for the conversations match the emotional condition of the previous round. Conversations are manually sorted into three emotional states: joyful, sad, and furious, with an additional category for others. The model and its modifications produce competitive outcomes when compared to the organisers' baseline and other participants.

Malte and Ratadiya [27] proposed a bi-directional transformer BERT architecture. They present a generative deep-learning solution based on the bidirectional transformer-based BERT architecture for cyber abuse detection across English, Hindi, and codemixed Hindi English (Hinglish) text. They utilise the dataset provided as part of the Trolling, Aggression, and Cyberbullying (TRAC-1) session at COLING 2018 as part of the Shared Task on Aggression Identification. They obtained an F1(Weighted) score of 0.6596 on the Hindi FB test set using BERT(multilingual cased) and 0.6425 on the English FB test set

Table 2 Comparative analysis: technique and limitations

S.No.	Authors	Technique & dataset used	Results	Limitations
1	Feng and Cheng [15]	A multi-channel convolutionary neural network (MCNN-MA) on Taobao Chinese review data and Songbo's Chinese hotel review data	Their model has a better classification precision and comparatively cheaper training time costs	Developed and tested in Chinese language only
2	Luo et al. [25]	Sequence Generative Adversarial Networks (SeqGAN) on SST and HT dataset	For enhancing the accurate analysis of Sentiment. Usability is increased by 24.6%, enhancing the classification precision by 1%	Classification accuracy does not increase significantly
3	Yin et al. [45]	Proposed FCP-Lex for The emotional lexicon's creation method on MRD & LMRD	The precision was 82%.	Specific to English Corpora only
4	Ahmad et al. [1]	Used transfer learning with cross-lingual embeddings and Showed knowledge transfer to other domain on Emo-Dis-HI data	53% F1 score	Didn't focus on contextual meaning
5	Badugu et al. [41]	Used NB and KNN classification techniques on Tweets	Achieved 72.06% accuracy using NB	Low utilisation of contextual meaning
6	Seal et al. [39]	Their own proposed algorithm on ISEAR dataset	Emotions were detected using phrasal verbs in particular	Inadequate vocabulary and No contextual information
7	Seo et al. [40]	Heterogeneous modality transfer learning (HMTL) on the CMU-MOSI and IEMOCAP dataset	In unimodal performances, The Baseline + HMTL model produce 62.6% accuracy	The sensitivity is dependent on the parameters of the training data Due to its numerous learning phases, it needs a longer learning time
8	Al-Azani and El-Alfy [2]	Two classifiers were used: (1) SVM (LibSVM) and Linear Kernel, and (2) logistic regression (L2 norm regularization and Liblinear solver) are applied to the self-developed SADAM dataset	Precise prediction of speech with a precise Emotion with 94.08% accuracy	Restricted to the Arabic language

Table 2 (continued)

S.No.	Authors	Technique & dataset used	Results	Limitations
9	Lu et al. [24]	RNN as a sentiment classifier with self-attention on SWBD-sentiment and IEMOCAP Data	70.10% on SWBD-sentiment and on IEMOCAP from 66.6% to 71.7%	Supervised learnt speech features are required
10	Zhang et al. [47]	Knowledge Guided Capsule Attention Network (KGCapsAN) on 6 Datasets	Upto 85.45% accuracy on Rest14 dataset	Need of further comprehensive knowledge for sentiment classification
11	Wang et al. [44]	used CNN-LSTM on four datasets: Stanford Sentiment Treebank, EmoBank, The Valence and Arousal Facebook Posts, Chinese Valence-Arousal Texts	The proposed Regional CNN-LSTM (Tree-structured) outperformed LSTM (HAN, two-layer Attention) with a significant statistically difference (p less than 0.05)	domain-specific
12	Huang et al. [18]	BERT on the EmotionLines dataset	F1 Score: 81.5% on Friends, and 88.5% for the EmotionPush data	Amount of data was insufficient
13	Huang et al. [19]	HRLCE and BERT model on Tweets	Achieved F1 Score of 77.9%	There were many misclassifications
14	Ragheb et al. [36]	Attention based model for emotions classification on SemEval-2019 Task 3 dataset	Achieved F1 Score: 75.82%	Does not function well in identifying happiness
15	Malte et al. [27]	BERT architecture on Facebook multilingual texts	F1 score: 55.20% on English, and 45.21% on Hindi texts	The system's effectiveness was hampered by the existence of slang in the dataset
16	Polignano et al. [34]	Compared the efficiency of the FastText, GloVe Embedding, and Google Word Embedding on the SemEval-2018 (Task1), ISEAR, and the SemEval-2019 (Task 3) data	FastText worked better	The proposed model is somewhat complicated
17	Singla et al. [21]	Used various ML techniques on self developed dataset, BHAAV, for Hindi	Logistic regression produces an accuracy of 62%	The precision achieved was only 58%. The model did not recognise classes except Neutral
18	Manshu and Bing [28]	A prior-knowledge information hierarchical attention network (HANP) on the Amazon reviews	Have 87.76% mean accuracy	Inadequate amount of data

using LSTM+augmented data. It has also been discovered that using transliteration as a preprocessing step boosts performance substantially on the Hindi test set.

Polignano et al. [34] presented a word embedding method for text-based emotion detection. The emotions experienced have shown to be quite essential in creating a full user profile that includes her psychological features. As a result, it is of particular significance to the research community to create ways for recognising emotions from the text that are accurate and robust in common writing circumstances. They opted to examine the model's outcomes acquired by altering three distinct pre-trained word embeddings: Google word embeddings (GoogleEmb) [29], GloVe (GloVeEmb) [33], and FastText (FastTextEmb) [20]. They used SemEval 2019 task 3, SemEval 2018 task 1, and ISEAR. [38] compiled the ISEAR dataset. It is made up of 7666 incidents that are annotated by seven basic emotions: joy, fear, rage, sadness, disgust, and guilt. For the SemEval 2018 task 1 dataset, our technique achieved a maximum F1 score of 0.84, compared to 0.78 for the SVM baseline. They attained a maximum F1 score of 0.703 for the SemEval 2019 task 3 dataset, compared to 0.44 for the Random Forest classifier.

Al-Azani and El-Alfy [2] used two classifiers for emotion recognition. Combining textual, aural, and visual modalities, an improved technique for video analytics is offered to predict the speaker's emotion in multi-dialect Arabic. To represent each modality, different characteristics are retrieved, such as prosodic and spectral acoustic features for audio, neural word embedding for audio text transcripts, and dense optical-flow descriptors for visual modality. They created a dataset from relevant opinion videos gathered on YouTube owing to the lack of accessible video datasets for multimodal Arabic sentiment analysis. This dataset is known as the Sentiment Analysis Dataset for Arabic Multimodal (SADAM). The proposed multi-level hybrid fusing feature level with score level achieved the greatest accuracy of 95.08%. It yielded outcomes that improved classification accuracy by more than 6% compared to the previous state of the art models.

Wang et al. [44] suggested the tree-structured regional model of CNN-LSTM. To predict the valence-arousal (VA) ratings of texts, this paper offers a tree-structured regional CNN-LSTM model composed of two parts: regional CNN and LSTM. Four emotional corpora were utilised in this experiment: Stanford Sentiment Treebank (SST) [32], EmoBank [8], The Valence and Arousal Facebook Posts (FB) [35], Chinese Valence-Arousal Texts (CVAT) [46]. On the testing dataset, the model has a Pearson correlation coefficient (r) of 0.809 on SST, 0.626-0.504 (valence-arousal) on EB (reader), 0.566-0.499 (valence-arousal) on FB, and 0.793-0.580 (valence-arousal) on CVAT.

Zhang et al. [47] proposed Knowledge Guided Capsule Attention Network (KGCapsAN). They suggested a capsule attention network and a Bi-LSTM network. The capsule attention network uses the attention mechanism to implement the routing strategy. Experiments are carried out using six datasets: the Twitter corpus, Rest15 (from SemEval-15 Task 12), Rest16 (from SemEval-16 Task 5), SpATSA, Lap14, and Rest14 (from SemEval-14 Task 4). They achieved accuracy of 74.28%, 76.28%, 81.96%, 81.12%, 73.18%, and 88.47% on Twitter, lap14, Res14, Res15, SpATSA and Res16 dataset respectively.

Lu et al. [24] used RNN as a sentiment classifier with self-attention. They proposed pre-training with end-to-end automated speech recognition (ASR) and solving the audio Sentiment as a downstream challenge. They generate the SWBD sentiment dataset. It comprises 140 hours of speech with 49,500 classified utterances, making it ten times larger than the current biggest, IEMOCAP . They enhance the state-of-the-art sentiment analysis accuracy on IEMOCAP from 66.6% to 71.7%. They outperformed strong baselines on SWBD-sentiment, achieving 70.10% accuracy on the test set.

Yin et al. [45] worked on the construction of a Feeling/Emotion vocabulary based on the part of speech chunks, specifically CP chunks and proposed an automatic construction method of the sentiment lexicon. They named this FCP-Lex. They notice that (1) the Sentiment Lexicon's surrounding words, with the same meaning, can be different, (2) the Part of Speech (POS) information of these surrounding words is likewise diverse, and (3) the varied POS of these surrounding words can be utilised to identify feelings. The varied POS of surrounding words is then used to create numerous CP-chunks for each sentimental word, and researchers give distinct sentimental values to these CP-chunks using the CHI method. The classification step follows them. They used two movie review datasets: the long movie review dataset (LMRD) [26], which contains 100000 sentences (25000 negative labelled, 25000 positive labelled, 50000 unlabeled sentences). Another movie review data (MRD) [31] is labelled corpora with 5331 positive and an equal number of negative labels. They achieved an accuracy of 71.34% and 82.10% on MRD and LMRD, respectively. Experiments revealed that FCP-Lex is more successful than other sentiment lexicons.

Ahmad et al. [1] proposed Borrow from rich cousin- transfer learning for emotion detection using cross-lingual embedding. They present an effective strategy for overcoming resource constraints in Hindi emotion recognition by using information from a resource-rich language such as English. From the ground up, They constructed a dataset called Emo-Dis-HI. They monitor famous Hindi news websites for disaster-related news documents (both natural and artificial). The baseline models are evaluated using two datasets, Emo-Dis-HI and EmoSemEval-EN. With a micro-average F1-Score of 0.863, this model performs well on Emo-SemEval-EN data. On their dataset, however, the F1 score is just 0.53.

Matla and Badugu [41] performed emotional analysis on Twitter data. They produced labelled data using a rule-based method in prior work, referencing Russell's Circumplex model. They utilised the supervised eager learning classifier (Naive Bayes) and supervised lazy learning classifier (k-nearest neighbour) algorithms. The sentiment140 dataset was collected. Although the dataset contains several properties such as a tweet, label, and so on, they were only interested in the text attribute, i.e., tweet. They achieved an accuracy of 72.60% by using Naïve Bayes and 55.50% using KNN.

Seal et al. [39] presents a practical emotion identification approach at the sentence level from the text by searching direct emotional terms from a pre-defined emotion keyword library. They used Python and C# programming language to implement their emotion detection algorithm. They utilised the 'ISEAR' standard extensive emotion dataset; around 7500 phrases in the dataset deal with anger, joy, sadness, fear, contempt, and guilt. The dataset is a collection of writings that have been accurately classified with emotion. They built an individual emotion dataset with emotion keywords that correspond to the individual feelings. The accuracy obtained was 65%, the Precision value was 70.11%, Recall value of 62.67%, and F1 score of 66.18%.

Seo et al. [40] proposed "Heterogeneous modality transfer learning" (HMTL). They proposed integrating knowledge from a source modality of textual data in transfer learning to improve audio, visual emotion and sentiment analysis performance. They use an adversarial neural network with a decoder and have three stages. In this situation, the target model learns from the source model and increases performance despite the model only using audio and visual input and no textual data. Two multimodal datasets, CMU-MOSI and IEMOCAP, were used. We can test a multinomial classification problem with these datasets because it contains nine emotions. The Baseline + HMTL model performs best on CMU-MOSI data, with 62.6% accuracy and 60.8% F1 score for the auditory data and 64.8% accuracy and 61.7% F1 score for the visual modality. Unimodal performance on the IEMOCAP was 51% for auditory data and 51.6% for visual data, respectively.

Feng and Yan [15] proposed a multi-channel convolutionary neural network (MCNN-MA). A unique sentiment analysis model (MCNN-MA) is developed based on a multi-channel convolutional neural network with a multi-head attention mechanism. The performance of the proposed model MCNN-MA is validated using two data sets in this research. Taobao Chinese review data and Songbo's Chinese hotel review dataset. The other data source is the Taobao Chinese review data set. Each data sample is classified with positive or negative labels in both data sets. Both sets of data are imbalanced. Tan Songbo's dataset includes 10,000 ratings (7,000 good + 3,000 bad). While Taobao dataset includes 18,875 feedback from customers (9,549 good + 9,326 bad). On the two data sets, the three-channel feature combination attained the highest accuracy (86.32%, 85.29%), demonstrating that the three-channel setup is both essential and effective.

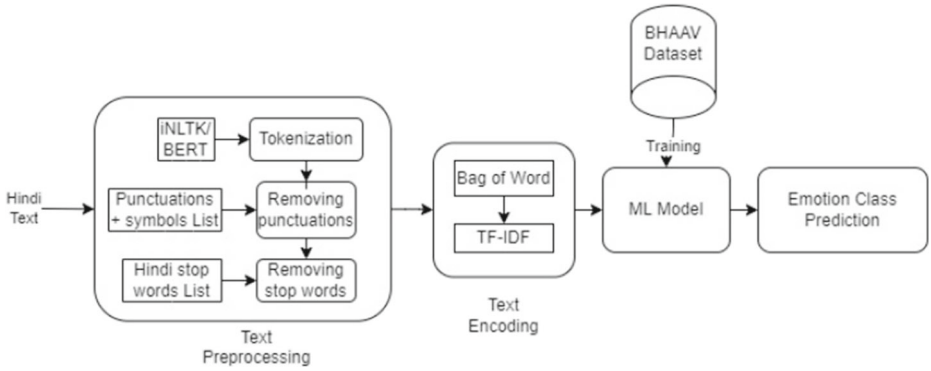
Luo et al. [25] proposed Sequence Generative Adversarial Networks (SeqGAN). SeqGAN is used to increase the quality of data and make it diverse. Long short-term memory (LSTM) networks with attention mechanisms compress sentences for SeqGAN training data. Stanford Sentiment Treebank (SST) and Hate Speech (HS) Dataset were used for experiments. In addition, they presented a data screening approach for obtaining more reliable data from created data. It provides an average of 80.1% and 75.6% accuracy on the SST and HS datasets, respectively.

The current research in the area of emotion recognition from text is mostly on identifying the polarity of emotion (positive/negative) and that also in the English Language as shown in Table 2. Identifying varieties of emotions and their intensity is one of the least investigated sentiment domains. As a result, sentiment analysis and emotion identification in languages other than English, particularly regional languages, present a significant challenge and potential for research.

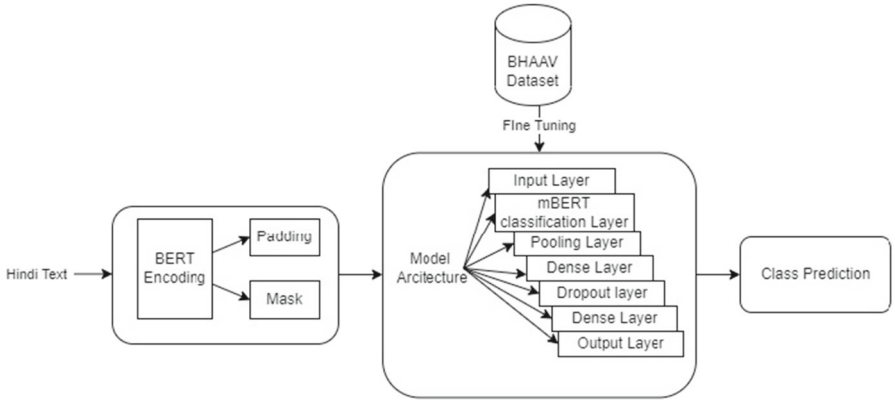
3 Emotion recognition framework

This research focuses on speech sentiment and emotion analysis, specifically for the Hindi language. The study includes three media dimensions: text, speech, and video. The iNLTK library, the deep learning library with Indian language support, is used for natural language processing. Figure 1 depicts the essential steps that are applied for emotion recognition. Figure 1 by dividing it into two subfigures, Fig. 1a and b. Figure 1(a) showcases the components level flow when the BHAAV dataset is trained and tested upon the state-of-the-art machine learning algorithms. Whereas figure 1(b) displays the component level flow when the BHAAV dataset is trained and tested upon mBERT.

- **Input File:** The input file refers to the file that contain the sentences to be classified in various emotion categories. The file is read by our python program and stored sentences wise in a list. And each sentence is extracted from the list one by one, and then fed to the text preprocesser.
- **Text Preprocessing:** This is the critical phase in the prediction model. We used the iNLTK library to preprocess the text. In this step, we perform tokenisation, removing punctuation marks, removing pronouns, and removing stop words. In machine learning algorithms, when we apply Tf-Idf, it focuses on the frequency of data, so when we perform text analysis on data such as comments or tweets, removing noise comes in handy. It improves the quality of data, and features are now highlighted.
- **Text Encoding:** Transforming words into numbers, and sequences of words into sequences of numbers, is the key to performing any text-based project, such as emotion



(a) Components level flow for machine learning algorithms



(b) Components level flow for mBERT architecture

Fig. 1 Framework for emotion recognition from hindi text

analysis. Once we have numbers, it is all about data analytics, and machine learning algorithms can assist us with classification and clustering. Text encoding uses Bag of Words (BOW), TF-IDF Encoding, Word2Vector Encoding, and BERT Encoding. We used all of these techniques in our experiment.

- **BHAAV Dataset:** In machine learning, a dataset is simply a collection of data pieces that can be treated as a single unit by a computer for analytic and prediction purposes. This means that the data gathered should be uniform and understandable to a machine, which does not perceive data in the same way that humans do. After collecting the data, it is critical to preprocess it by text cleaning and to prepare it, annotating the data, and providing constructive computer-readable tags. We trained our model on the BHAAV dataset and then applied it to the emotion prediction task.
- **Machine Learning Model:** The research compares various machine learning and deep learning models such as Naive Bayes, Logistic Regression, SVC classifier, Decision Tress classifier, Boosted Trees, KNN classifier, Random Forest, OvA classifier, as well as the BERT multilingual transformer, on various word embedding techniques.

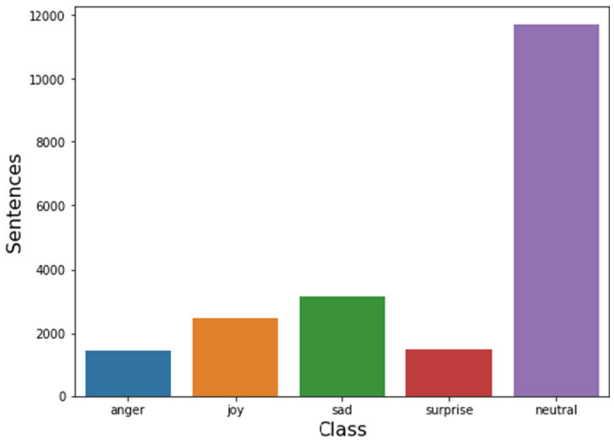


Fig. 2 Sentence distribution in class

- **Emotion Class Prediction:** Emotion detection is a text classification task. We used text classification models and trained them on the BHAAV dataset. Now the model can predict the class of input data fed to it by the users.

4 BHAAV dataset

BHAAV is India's first and most significant collection of Hindi text for analysing emotions expressed by a writer's story's characters as perceived by a reader. This dataset contains 20304 sentences in the Hindi language, chosen from 230 short tales in 18 different categories, including Inspirational and Mystery. Every sentence has been labelled with five emotion categories: joy, suspense, anger, sadness, or neutral. The dataset was manually annotated by annotators that were native Hindi-speaking volunteers with at least ten years of formal education in Hindi and a strong interest in reading the stories. There were five significant categories of emotions prevalent in the corpus: anger, joy, suspense, sadness, and neutral. Sentence distribution among various classes is shown in Fig. 2.
It can be clearly seen that the original dataset is highly imbalanced and contains approx 60% neutral class only. So before applying the machine learning algorithm, we have to balance the class in the dataset. Because the majority of machine learning algorithms perform best when the number of instances or samples in each class is nearly equal, it is because to the fact that most of the algorithms are designed to maximise accuracy while minimising errors. However, if the data set is imbalanced, users can achieve a high level of accuracy simply by predicting the majority class, but they will miss out on capturing the minority class, which is frequently the point of creating the model in the first place.

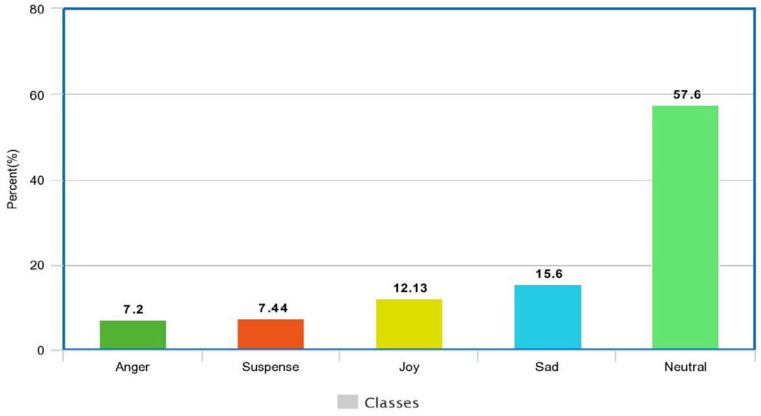
SMOTE technique can be used to generate as many synthetic examples as needed for the minority class. As described in the paper [12], it is suggested that random undersampling be used first to reduce the number of examples in the majority class, followed by SMOTE to oversample the minority class to balance the class distribution. Figure 3 shows the class distribution in the dataset. Figure 3a shows distribution before SMOTE, while Fig. 3b shows distribution after SMOTE. From Fig. 3b, it is shown that SMOTE was not effective in the case of our dataset. So we manually over-sampled the minority classes in order to balance

the dataset. Figure 4 is the final dataset after class balancing, and this dataset is used for our experiments.

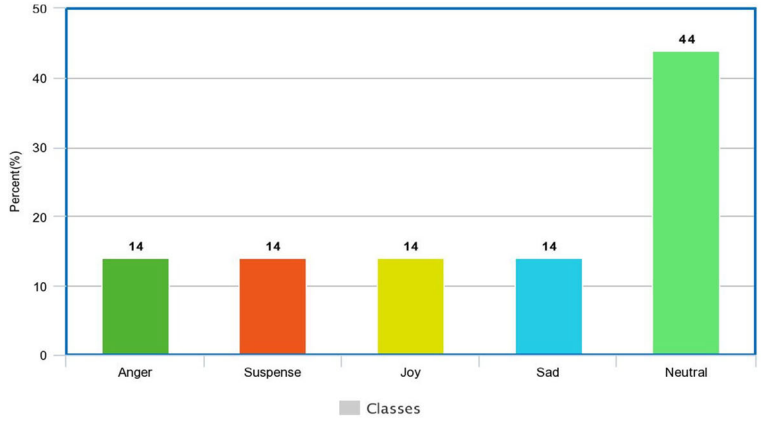
Table 3 represents the summary of various datasets used till date for detection of emotions. It can be observed that BHAAV is one of its kind dataset available for Hindi language. The datset provides sufficient number of labeled hindi sentences to train and test a model.

5 Emotion recognition using ML classifiers with TF-IDF approach

1. **Naive Bayes Classifier:** Figure 5a shows the confusion matrix of this model on the test data.
   Accuracy :55.15%, Precision(weighted):61.89%, Recall(weighted):55.15%, and F1 score(weighted):53.08%.
2. **Logistic Regression Classifier:** Figure 5b shows the confusion matrix of this model on the test data. Accuracy :74.09%, Precision(weighted):73.49%, Recall(weighted):74.09%, and F1 score(weighted):73.66%.
3. **SVC Classifier:** Figure 5c shows the confusion matrix of this model on the test data.



(a) Before SMOTE



(b) After SMOTE

Fig. 3 Dataset balancing using SMOTE

Table 3 Summary of various datasets for emotion detection

S.No.	Dataset	Content & Language	Features	Emotion model
1	ISEAR [38]	7665 labeled English sentence	7 emotion labels	Discrete
2	SemEval [37]	1250 Arabic and English texts obtained from Tweets, and major newspapers	6 emotion labels	Discrete
3	EMOBANK [7]	10000 labeled English sentence, spanning over a wider domain	6 emotion labels	Dimensional
4	WASSA-2017 EmoInt [30]	Annotated tweets	4 emotion labels	Discrete
5	Cecilia Ovesdotter Alm's Affect [4]	Tales	6 emotion labels	Discrete
6	Daily Dialog [22]	13118 labeled sentence	7 emotion labels	Discrete
7	CrowdFlower [5]	39740 labeled tweets	13 emotion labels	Discrete
8	Grounded emotions [23]	2557 tweets	2 emotion labels	Discrete
9	Emotion-Stimulus [16]	1594 labeled sentence	7 emotion labels	Discrete
10	The Valence and Arousal data [35]	2895 Facebook Posts	Valence and Arousal labels	Dimensional
11	MELD [38]	Over 1400 English dialogues and 13000 utterances from TV Show 'Friends'	7 emotion labels	Discrete
12	Emotion Lines [17]	2000 English dialogues and 29245 labeled utterances from 'FRIENDS' and FB chats	7 emotion labels	Discrete
13	Smile dataset [43]	3085 tweets	5 emotion labels	Discrete
14	AMAN'S Emotion dataset [11]	1446 labeled sentence	7 emotion labels	Discrete
15	BHAAV dataset [21]	20304 labeled Hindi sentence	5 emotion labels	Discrete

Accuracy :78.70%, Precision(weighted):78.16%, Recall(weighted):78.70%, and F1 score(weighted):78.30%.

4. **Decision Trees Classifier:** Figure 5d shows the confusion matrix of this model on the test data.
   Accuracy :23.62%. Precision(weighted):30.36%, Recall(weighted):23.62%, and F1 score(weighted):11.07%.
5. **Boosted Trees Classifier:** Figure 5e shows the confusion matrix of this model on the test data.
   Accuracy : 31.03%, Precision(weighted):35.03%, Recall(weighted):31.03%, and F1 score(weighted):29.97%.
6. **KNN Classifier:** Figure 5f shows the confusion matrix of this model on the test data.
   Accuracy : 63.94%, Precision(weighted): 65.68%, Recall(weighted): 63.94%, and F1 score(weighted): 60.68%.
7. **Random Forest Classifier:** Figure 5g shows the confusion matrix of this model on the test data.
   Accuracy :81.26%, Precision(weighted):81.08%, Recall(weighted):81.26%, and F1 score(weighted):81.12%.
8. **One Vs All Classifier:** Figure 5h shows the confusion matrix of this model on the test data.
   Accuracy :71.00%, Precision(weighted):72.00%, Recall(weighted):71.00%, and F1 score(weighted):68.00%.

6 Emotion recognition using BERT transformer

BERT is an acronym for "Bi-directional Encoder Representations from Transformers". It is a Google-developed transformer-based machine learning approach for pre-training natural language processing (NLP). Jacob Devlin and his Google colleagues created and published

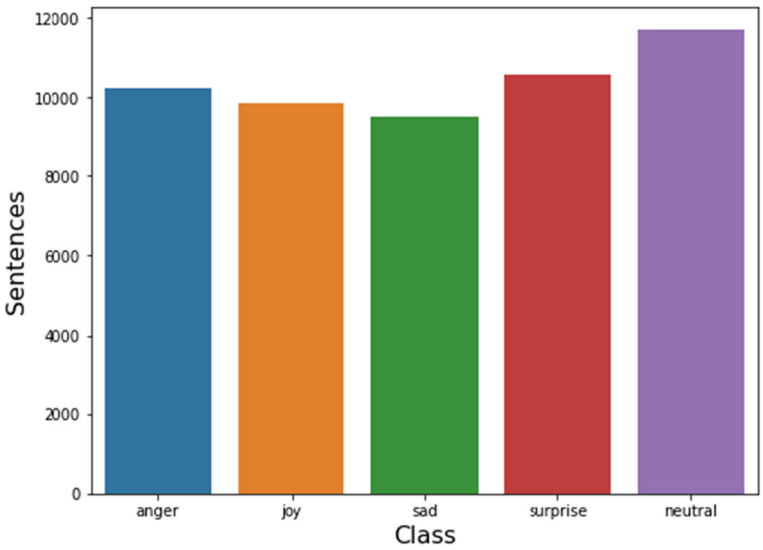


Fig. 4 Dataset balancing using manual class weight

BERT in 2018. Google announced in 2019 that It had started using BERT in its search engine, and by late 2020, it had integrated BERT into nearly every English-language query. According to a 2020 literature review, BERT has become a ubiquitous baseline in NLP studies in little more than a year, with over 150 research articles analysing and developing the model.

The biggest challenge in NLP was creating a large enough dataset to train a model to understand a language. Training a language is a difficult task, and BERT has been trained on trillions of wiki pages, making it extremely powerful in understanding the context of any sentence. Bert is free to use, easily fine-tuned, and simple to implement.



(a) Naive Bayes Classifier (b) Logistic Regression (c) SVC Classifier

(d) Decision Trees (e) Boosted Trees (f) KNN classifier

(g) Random Forest (h) One Vs All Classifier

Fig. 5 Confusion matrix of various state-of-the-art ML algorithms tested on the BHAAV dataset

6.1 Bert architecture

The BERT was originally developed for English Language and that version is available in two sizes: BERT_BASE, and the BERT_LARGE. Both models are pre-trained with data from BooksCorpus [48], which contains 800 million unlabeled words, and English Wikipedia, which contains 2500 million words. There are 12 encoder in BERT Base and 24 in case of BERT Large.

A Basic Transformer comprises an encoder (which reads the input) and a decoder (which produces prediction). BERT only employs the encoder component, as shown in Fig. 6. BERT's encoder takes the tokens sequence as input after converting the tokens into some vector form.

BERT-Base, BERT-Large, BERT-Base, Multilingual, and BERT-Base Chinese are the available version of BERT. Each version is available in two versions, Cased and Uncased, having 12 to 24 encoders. In our model, we used mBERT. mBERT is a "multilingual cased BERT" model which is pre-trained on 104 popular languages, Hindi is one of them [14].

6.2 Bert model summary

Before feeding data to the *bert_classification* layer in BERT model, we first have to encode the data into tensor. Bert *tokenizer* convert the text into tensor having attributes *input_ids* and *attention_mask*. *tokenizer* function also add padding to the sentences that have less word the

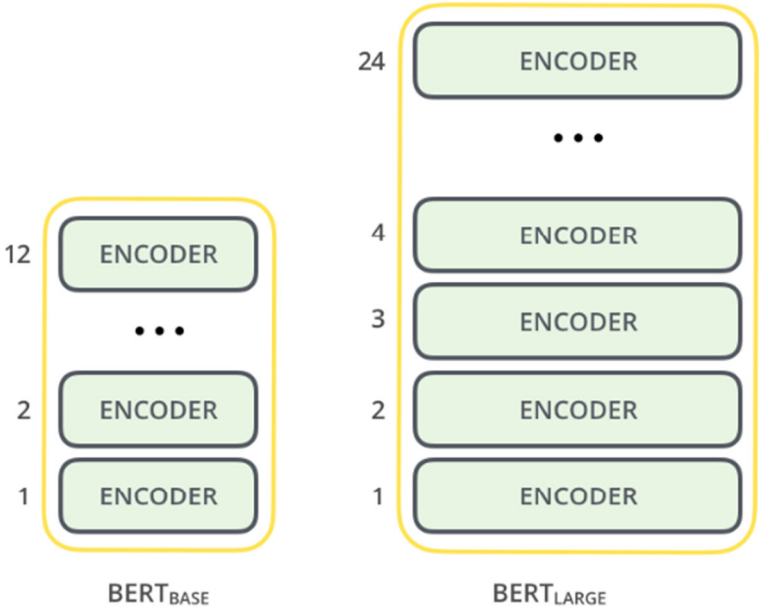


Fig. 6 Bert architecture. Source: [3]

```
Model: "model_2"
_________________________________________________________________________________________________
Layer (type)                 Output Shape              Param #     Connected to
=================================================================================================
input_ids (InputLayer)       [(None, 128)]             0
_________________________________________________________________________________________________
attention_mask (InputLayer)  [(None, 128)]             0
_________________________________________________________________________________________________
tf_bert_model_1 (TFBertModel) multiple                 177853440   input_ids[0][0]
                                                                   attention_mask[0][0]
_________________________________________________________________________________________________
global_max_pooling1d_2 (GlobalM (None, 768)            0           tf_bert_model_1[2][0]
_________________________________________________________________________________________________
dense_6 (Dense)              (None, 128)               98432       global_max_pooling1d_2[0][0]
_________________________________________________________________________________________________
dropout_76 (Dropout)         (None, 128)               0           dense_6[0][0]
_________________________________________________________________________________________________
dense_7 (Dense)              (None, 32)                4128        dropout_76[0][0]
_________________________________________________________________________________________________
dense_8 (Dense)              (None, 5)                 165         dense_7[0][0]
=================================================================================================
Total params: 177,956,165
Trainable params: 177,956,165
Non-trainable params: 0
_________________________________________________________________________________________________
```

Fig. 7 Bert model summary

max_padding_length, ensuring every tensor to be equal and *max_padding_length* in length. *input_ids* and *attention_mask* are then fed to the *bert_classification* layer.
As shown in the Fig. 7, there is one input layer having parameters *input_ids* and *attention_mask* in our mBERT model. The input layer is then fed to *TFBertModel1* layer and which is the input of *global_max_pooling1d_2* layer. Then this output is the input to the *dense_6* layer and which is input to the *dropout_76* layer and one more hidden layer *dense_7* and then the output of final layer is input to the output layer. The output layer returns a list containing emotions.

6.3 mBert model performance

On testing data, the performance of our model is: Balanced_Accuracy :91.84%, Precision:92.01%, Recall:91.74%, Loss:0.3073, and AUC:98.46%. The detailed model performance is shown in Fig. 8, where we compared the various validation and testing parameters.

Table 4 Comparison table for various models

S.No.	Model Name	Accuracy	Precision(W.)	Recall(W.)	F1(W.)
1	Naive bayes	55.15%	61.89%	55.15%	53.08%
2	Logistic regression	74.09%	73.49%	74.09%	73.66%
3	SVC classifier	78.70%	78.16%	78.70%	78.30%
4	Decision trees	23.62%	30.36%	23.62%	11.07%
5	Boosted trees	31.03%	35.03%	31.03%	29.97%
6	KNN classifier	63.94%	65.68%	63.94%	60.68%
7	Random forest	81.26%	81.08%	81.26%	81.12%
8	One Vs all classifier	71.00%	72.00%	71.00%	68.00%
9	CNN+Muril	59.38%	59.38%	57.56%	58.45%
10	**mBERT**	**91.84%**	**92.01%**	**91.74%**	**91.87%**

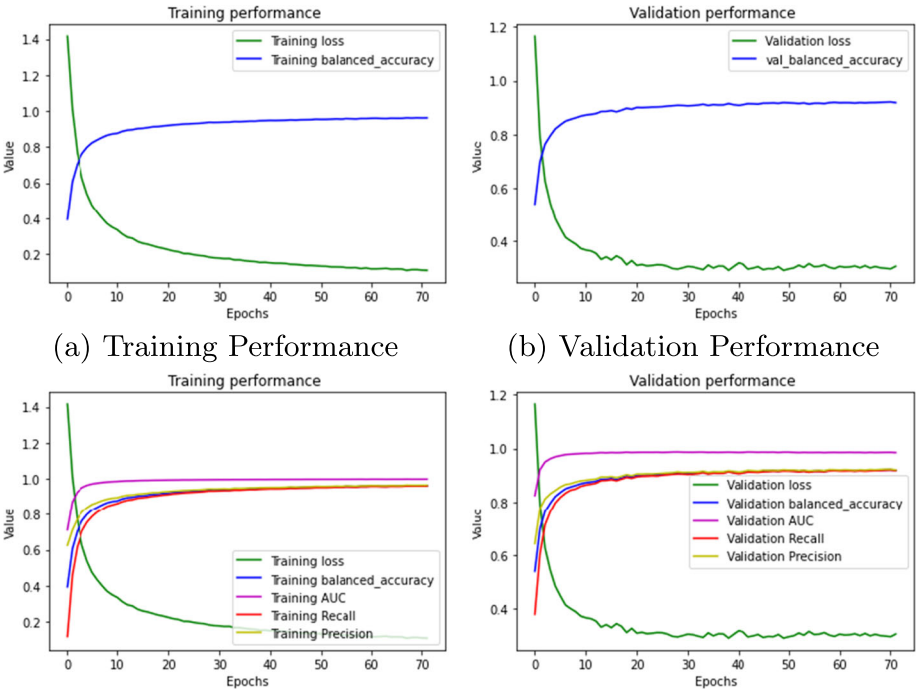
7 Results and discussions

Furthermore, our proposed mBERT model (highlighted in bold in Table 4) has achieved state-of-the-art performance on the dataset in emotion classification with an accuracy of 91.84% on the test data.

Figure 9 shows the bar graph for the accuracy of our various defined models. Further on comparing the best accuracy of the best performing model in paper [21], the plot is given in Fig. 10, which shows that our model outperformed their baseline models.

8 Implications and predictions

Word2Vec models generate context-independent embeddings, which means that each word has only one vector (numeric) representation. The word's various meanings (if any) are combined into a single vector. However, depending on the context, the BERT model generates embeddings with multiple (multiple) tensor (vector/numeric) representations for the



(a) Training Performance

(b) Validation Performance

(c) Training Performance (All Parameters)

(d) Validation Performance (All Parameters)

Fig. 8 mBERT model performance

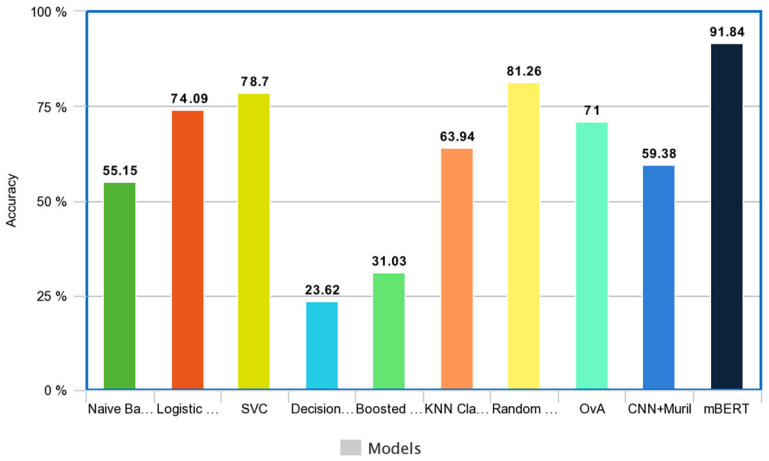


Fig. 9 Accuracy comparison of our models

same word. BERT embeddings are thus context-dependent. Thus BERT handles the word sense disambiguation problem.
Figure 11a, b, c, d and e shows the emotion class Sad, Angry, Surprise, Joy, and Neutral respectively.

9 Conclusion & future work

Emotion recognition from text is a fascinating problem with numerous dimensions of e-Learning, market research, social media analysis, genre predictions etc. This research investigates the challenges of emotion recognition and proposes a framework for emotions and sentiments detection in Hindi Language. mBERT Transformer is used over BHAAV

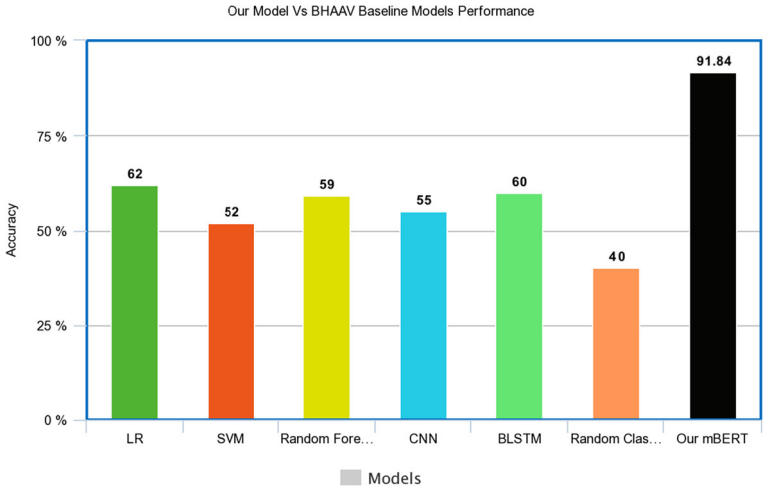
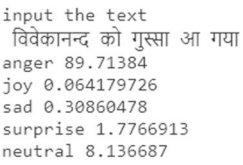
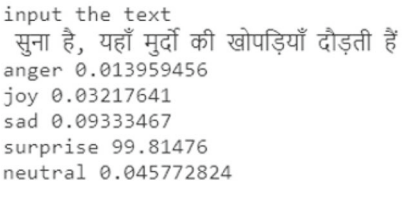


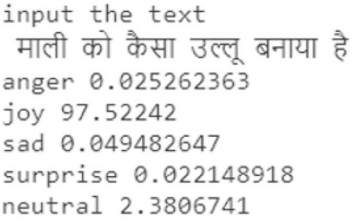
Fig. 10 Comparison of our mBERT with BHAAV models



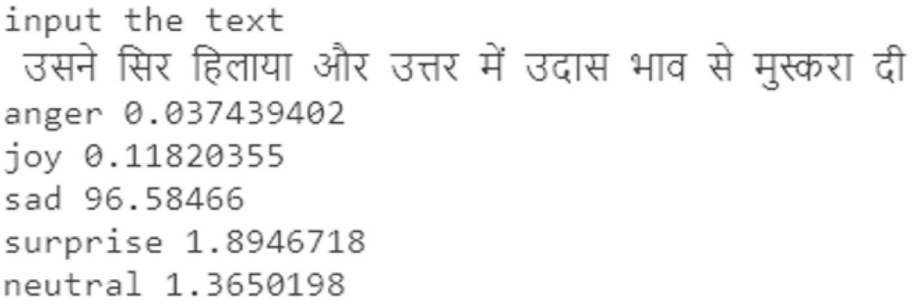
(a) Angry



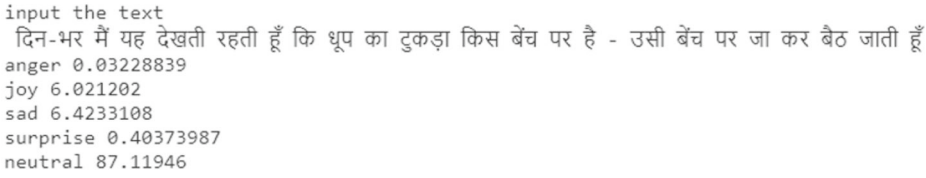
(b) Surprise



(c) Joy



(d) Sad



(e) Neutral

Fig. 11 Model predictions on emotion class

dataset to efficiently categorize the identified emotions. The results depicted that the model outperformed the other state-of-the-art techniques for emotion detection over same dataset with a balanced accuracy of 91.84%

The work presented in this research can be expanded in several ways. One of the tasks is to investigate the relationship between emotion categories and intensity. Since punctuation in chatting has increased, the relationship between punctuation and emotion classes can be examined. According to a recent study, there are 27 different types of emotions. Increasing

the number of classes will thus aid in better emotion classification. Cross-lingual emotion classification will be another area of future research. Cognitive and behavioural traits of speech are another upcoming features for designing an intelligent emotion recognition system.

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Data Availability The dataset analysed during the current study is now openly available on https://zenodo.org/record/3457467#.YuyoTXZBxPa Zenodo repository.

Declarations

Conflict of Interests The authors declare that there is no Conflicts of interests or Competing interests in this research.

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